9 Databases

In this chapter, you will learn about:
- ★ the design and use of single table databases
  - fields
  - records
  - validation
- ★ the use of basic data types
  - text/alphanumeric
  - character
  - Boolean
  - integer
  - real
  - date/time
- ★ the identification, purpose and use of primary keys
- ★ SQL scripts, used to query data stored in a single table database.

9.1 Databases

9.1.1 Single-table databases

A **database** is a structured collection of data that allows people to extract information in a way that meets their needs. The data can include text, numbers, pictures; anything that can be stored in a computer. Relational databases will be studied at A Level but for IGCSE only single-table databases will be studied.
A **single-table database** contains only one table.

Why are databases useful?

Databases prevent problems occurring because:

- if any changes or additions are made it only has to be done once – data is consistent
- the same data is used by everyone
- data is only stored once in relational databases which means no data duplication.

What are databases used for?

To store information about people, for instance:

- patients in a hospital
- pupils at a school.

To store information about things, for instance:

- cars to be sold
- books in a library.

To store information about events, for instance:

- hotel bookings
- results of races.

Find out more

Find five more uses for databases, and for each one decide what sort of information is being stored.

Fields and records – the building blocks for any database

Inside a database, data is stored in **tables**, which consists of many **records**. Each record consists of several **fields**. The number of records in a table will vary as new records can be added and deleted from a table as required. The number of fields in a table is fixed so each record contains the same number of fields.

An easy way to remember this is: each record is a row in the table and each field is a column in the table.

Note: while databases can contain multiple tables, all the databases considered in this chapter will contain a single table.

Table

Record 1	Field 1	Field 2	Field 3	Field 4
Record 2	Field 1	Field 2	Field 3	Field 4
Record 3	Field 1	Field 2	Field 3	Field 4
Record 4	Field 1	Field 2	Field 3	Field 4
Record 5	Field 1	Field 2	Field 3	Field 4
Record 6	Field 1	Field 2	Field 3	Field 4

Row

Column

▲ **Figure 9.1** Structure of a database table

A table contains data about one type of item or person or event, and will be given a meaningful name, for example:

- a table of patients called **PATIENT**
- a table of books called **BOOK**
- a table of doctor's appointments called **APPOINTMENT**.

Each record within a table contains data about a single item, person or event, for example:

- Winnie Sing (a hospital patient)
- IGCSE Computer Science (a book)
- 15:45 on January 2020 (an appointment).

As every record contains the same number of fields, each field in a record contains a specific piece of information about the single item, person or event stored in that record. Each field will have a meaningful name to identify the data stored in it, for example:

- For a hospital patient the fields could include:
  - The patient's first name field called **FirstName**
  - The patient's family name field called **FamilyName**
  - The patient's date of admission field called **DateOfAdmission**

- The name of the patient's consultant field called **`Consultant`**
- The patient's ward number field called **`WardNumber`**
- The patient's bed number field called **`BedNumber`**, etc.

The PATIENT table structure could look like this:

PATIENT Table

Record 1	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 2	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 3	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 4	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 5	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 6	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber

▲ **Figure 9.2** Structure of the PATIENT table

- For the table called BOOK the fields could include:
  - Title of the book called **`Title`**
  - Author of the book called **`Author`**
  - **`ISBN`**, etc.

Activity 9.1

State what fields would you expect to find in each record for the doctor's appointments and give each field a suitable name.

Note: Field names should be a single word, which should not contain any spaces, for example: **`BedNumber`**

Validation

Link

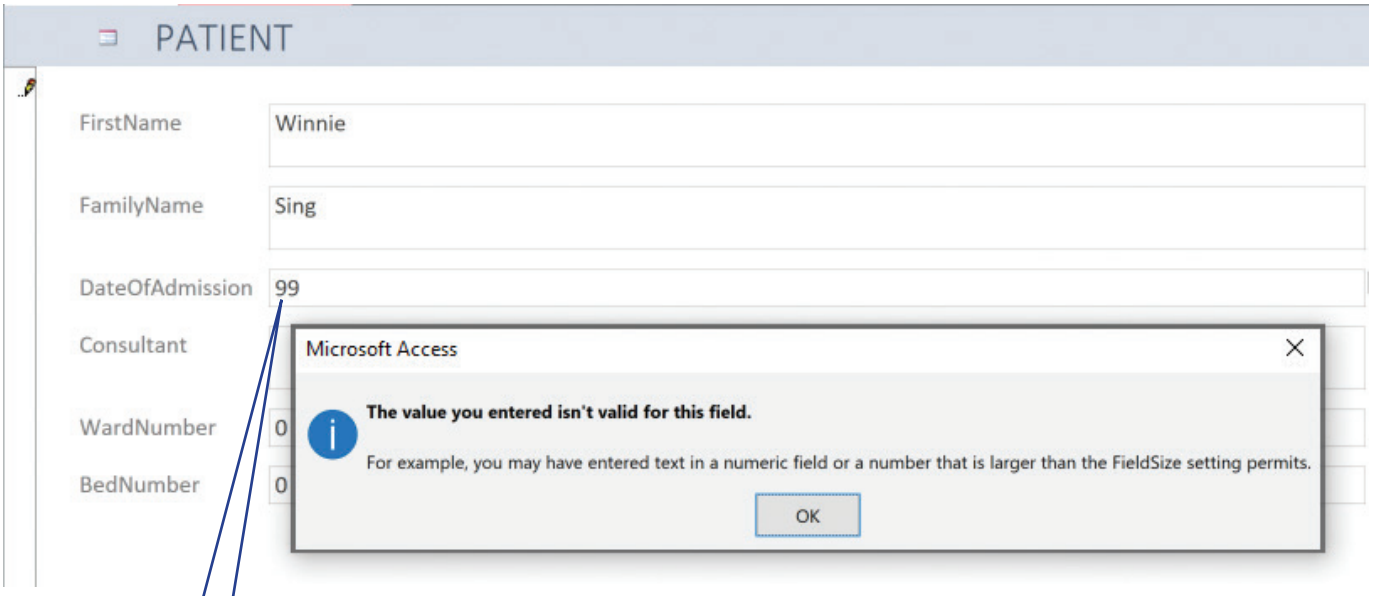
For more on validation see Section 7.5.

The role of validation was discussed in Section 7.5. It may be worth the reader revisiting this part of the book before continuing with this chapter.

Some validation checks will be automatically provided by the database management software that is used to construct and maintain the database. Other validation checks need to be set up by the database developer during the construction of the database.

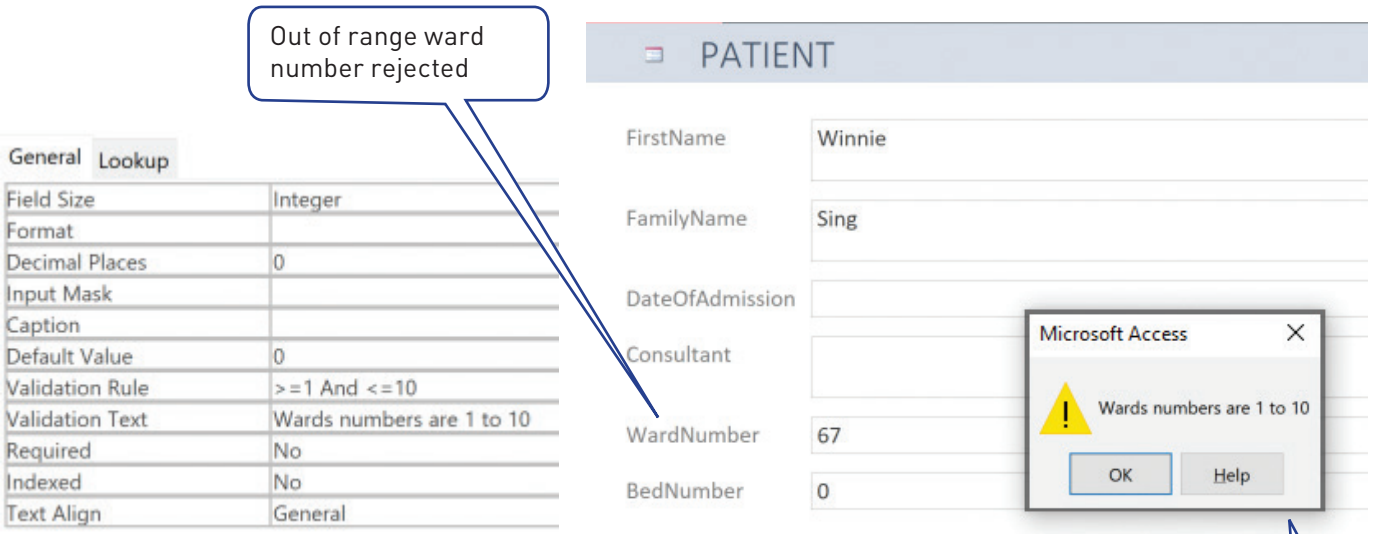
The practical use of a database management system is strongly recommended for all students. Practical examples will be used throughout this chapter. The database management software used is *Microsoft Access 365* as Microsoft Access is used by most schools with students studying IGCSE Computer Science.

For example, the **`DateOfAdmission`** field will automatically be checked by the software to make sure that any data input is a valid date before it can be stored in the PATIENT table.



▲ **Figure 9.3** Automatic validation for entering `DateOfAdmission` in the PATIENT table

However, the `WardNumber` field validation needs to be set up to allow only values 1 to 10 to be entered. This task needs to be completed by the database developer before the database is used.



▲ **Figure 9.4** Validation rule for entering `WardNumber` in the PATIENT table

9.1.2 Basic data types

There are six basic data types that you need to be able to use in a database:

- text/alphanumeric
- character
- Boolean
- integer
- real
- date/time.

What is a data type?

Each field will require a **data type** to be selected. A data type classifies how the data is stored, displayed and the operations that can be performed on the stored value. For example, a field with an integer data type is stored and displayed as a whole number and the value stored can be used in calculations.

These database data types are specified in the syllabus. They are available to use as *Access* data types, but the names *Access* uses may be different from the terms in the syllabus.

Syllabus data type	Description	Access data type
text/alphanumeric	A number of characters	short text/long text
character	A single character	short text with a field size of one
Boolean	One of two values: either True or False, 1 or 0, Yes or No	Yes/No
integer	Whole number	number formatted as fixed with zero decimal places
real	A decimal number	number formatted as decimal
date/time	Date and/or time	Date/Time

Find out more

Using Access (or any other suitable database management software) find all the different date and time formats that are available to use.

Activity 9.2

Using Access (or any other suitable database management software), set up a single table database to store the PATIENT records using the fields given previously:

`FirstName`, **`FamilyName`**, **`DateOfAdmission`**, **`Consultant`**, **`WardNumber`**, **`BedNumber`**.

- Choose a suitable data type for each field.
- Include validation checks to make sure that the ward number has whole number values from 1 to 10, and that the bed number has whole number values from 1 to 8.
- Enter the following data in the PATIENT table:

Winnie Sing, 12/10/2022, Mr Smith, 6, 8

Steve Chow, 23/10/2022, Miss Abebe, 6, 3

Chin Wee, 30/10/2022, Mr Jones, 7, 1

Min Hoo, 1/11/2022, Mr Smith, 6, 1

Peter Patel, 12/11/2022, Mr Jones, 7, 8

Sue Sands, 19/11/2022, Miss Abebe, 6, 2

Farouk Khan, 22/11/2022, Mr Jones, 7, 4

Ahmad Teo 22/11/2022, Mr Jones, 7, 2

You may want to look at the instructions on how to set up the CubScout database later in this chapter if you have not set up a database before.

9.1.3 Primary keys

As each record within a table contains data about a single item, person, or event, it is important to be able to uniquely identify this item. In order to reliably identify an item from the data stored about it in a record there needs to be a field that uniquely identifies the item. This field is called the **primary key**.

A field that is a primary key must contain data values that are never repeated in the table.

The primary key can be a field that is already used, provided it is unique, for example the ISBN in the book table. The PATIENT table would need an extra field for each record as all of the existing fields could contain repeated data. To create a primary key, we could add a new field to each record, for example a unique number could be added to each patient's record. The extra field is:

- Primary key field called **`HospitalNumber`**

Activity 9.3

Using *Access* (or any other suitable database management software), add the **`HospitalNumber`** field to the single table database PATIENT.

- Choose 'text' as the data type for this field.
- Include validation checks to ensure that 8 characters must be entered starting with **`HN`** followed by **`6 digits`** for example **`HN123456`**.
- Choose suitable data to store in each primary key field for the 6 patients in the table.

Activity 9.4

1. Write down **four** database data types. Describe each data type and give an example of a field where it would be a suitable choice.
2. Choose suitable words/phrases from the following list to correctly complete the paragraph that follows:

Word list:	database	fields	records
	record	table	primary key
	data type	validation	text
	char	Boolean	field

You will need to use some words more than once.

A single .... contains one ..... .

Each ..... consists of many ..... . Every .....

has the same number of ..... . Every ..... is given a

..... . Examples of data types are ....., ...

and ..... . Some ..... will have ..... rules.

9.1.4 SQL

Structured Query Language (SQL) is the standard query language for writing scripts to obtain useful information from a database. We will be using SQL to obtain information from single-table databases. This will provide a basic understanding of how to obtain and display only the information required from a database. SQL is pronounced as es-queue-el.

For example, somebody needing to visit a patient would only require the ward number and the bed number of that patient in order to find where they are in the hospital. Whereas a consultant could need a list of the names of all the patients that they care for.

SQL scripts

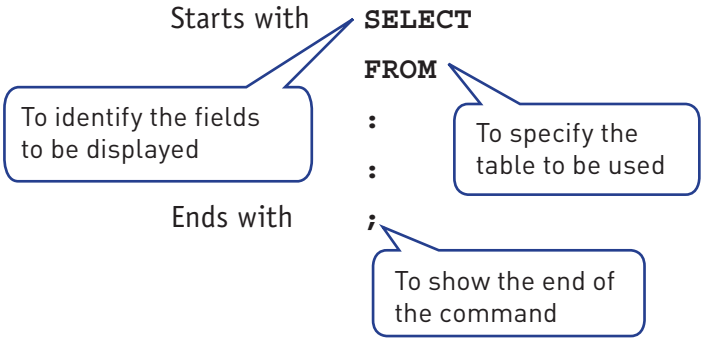
An **SQL script** is a list of SQL commands that perform a given task, often stored in a file so the script can be reused.

In order to be able to understand SQL and identify the output from an SQL script, you should have practical experience of writing SQL scripts. You can write scripts using SQL commands in *Access*. There are many other applications that also allow you to do this – *MySQL* and *SQLite* are freely available ones. When using any SQL application, it is important that you check the commands available to use as these may differ slightly from those listed in the syllabus and shown below.

You will need to be able to understand and identify the output from the following SQL statements.

SQL Query Statement	Description
`SELECT`	Fetches specified fields (columns) from a table; queries always begin with `SELECT`.
`FROM`	Identifies the table to use.
`WHERE`	Includes only records (rows) in a query that match a given condition.
`ORDER BY`	Sorts the results from a query by a given column either alphabetically or numerically.
`SUM`	Returns the sum of all the values in a field (column). Used with `SELECT`.
`COUNT`	Counts the number of records (rows) where the field (column) matches a specified condition. Used with `SELECT`.

An SQL command:



Only the `SELECT` and `FROM` commands are mandatory in an SQL script. All other commands are optional.

A **SELECT** statement takes the form:

`SELECT Field1, Field2, Field3`, etc. – this specifies the individual fields (columns) to be shown.

`SELECT *` – this specifies that **all** fields (columns) are to be shown.

A **FROM** statement takes the form:

FROM TableName – this specifies the table to use.

A **WHERE** statement takes the form:

WHERE Condition – this specifies the condition to apply.

Conditions often include values from fields, these values need to be stated in a form that matches the data type for the field.

Field type	Example value	General notes	Access notes
text	'Mr Smith'	Text field values should be in enclosed in single quotation marks.	Double quotation marks can also be used.
character	'M'	Character field values should be in enclosed in single quotation marks.	Double quotation marks can also be used.
Boolean	TRUE	Boolean can be **TRUE** or **FALSE**	Data type is Yes/No
integer	12	Integer field values should be whole numbers.	Allows integer or decimal values.
real	12.01	Real field values should be decimal numbers.	Allows integer or decimal values.
Date/time	'22/11/2022'	Date/time field values should be in enclosed in single quotation marks.	Date/time field values **must** be in enclosed in hashes (**#**).

Conditions also require operators to compare values from fields.

Operator	Description
=	equal to
>	greater than
<	less than
>=	greater than or equal to
<=	less than equal to
<>	not equal to
BETWEEN	between a range of two values
LIKE	search for a pattern
IN	specify multiple values
AND	specify multiple conditions that must all be true
OR	specify multiple conditions where one or more conditions must be true
NOT	specify a condition that must be false

An **ORDER BY** statement takes the form:

ORDER BY Field1, Field2, etc. – this specifies a sort in ascending or alphabetical order starting with the first field.

ORDER BY Field1, Field2 DESC – this specifies a sort in descending or reverse alphabetical order starting with the first field.

A **SUM** statement takes the form:

SELECT SUM (Field) – this specifies the field (column) for the calculation. The field should be integer or real.

A **COUNT** statement takes the form:

SELECT COUNT (Field) – this specifies the field (column) to count if the given criterium is met.

Example 1: Display consultant's patients

For example, the following SQL command for the PATIENT single-table database would provide a list of all Mr Smith's patients showing the hospital number, first name and family name for each of his patients.

```
SELECT HospitalNumber, FirstName, FamilyName
FROM PATIENT
WHERE Consultant = 'Mr Smith';
```

would display:

HospitalNumber	FirstName	FamilyName
HN123456	Winnie	Sing
HN123458	Min	Hoo

▲ **Figure 9.5** Output from the PATIENT table showing Mr Smith's patients

Example 2: Display consultant's patients in alphabetical order

This SQL command sorts the records in alphabetical order of family name:

```
SELECT HospitalNumber, FirstName, FamilyName
FROM PATIENT
WHERE Consultant = 'Mr Smith'
ORDER BY FamilyName;
```

would display:

HospitalNumber	FirstName	FamilyName
HN123458	Min	Hoo
HN123456	Winnie	Sing

▲ **Figure 9.6** Output from the PATIENT table showing Mr Smith's patients in alphabetical order of family name

Activity 9.5

1 Using the single table database PATIENT you have created.
   a Write an SQL query to list all Mr Jones' patients.
   b Write an SQL query to list all the patients not in ward 6.
   c Write an SQL query to list all the patients who arrived on 12/11/2022.
   d Write an SQL query to list all the patients who arrived between 12/10/2022 **AND** 30/10/2022.
2 Write down the output from this SQL query.

```
SELECT FirstName, FamilyName, BedNumber
FROM PATIENT
WHERE WardNumber = 7;
```

Activity 9.6

1 Using the single table database PATIENT you have created.
   a Write an SQL query to count the number of patients in ward 7.
   b Write an SQL query to count the number of patients not in ward 7.
2 Write down the output from this SQL query.

```
SELECT HospitalNumber, FirstName, FamilyName, Consultant
FROM PATIENT
ORDER BY Consultant, FamilyName;
```

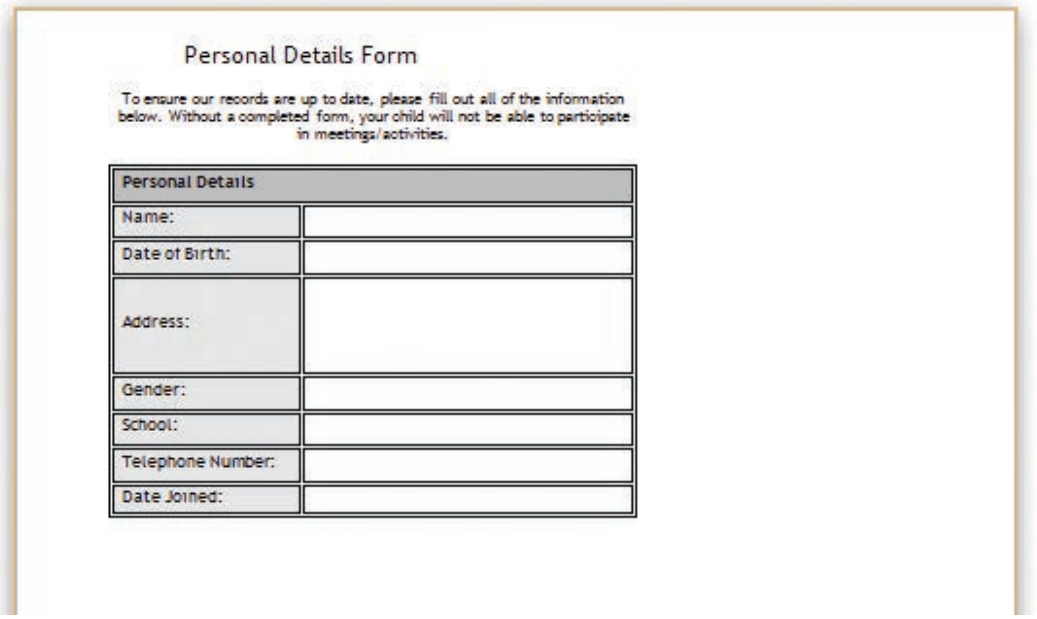
Practical use of a database

As an IGCSE Computer Science student you need to be able to do the following:

- define a single-table database from given data storage requirements
- choose a suitable primary key for a database table
- read, complete and understand SQL scripts.

In order to do this, you will need to use a database management system. The following case study shows how to set up a database with *Microsoft Access 365* and complete the tasks described above.

Boys and girls between the ages of seven and eleven can join a cub scout group. (http://en.wikipedia.org/wiki/Cub_Scout). Each cub scout group needs to keep records about its members. Most groups will keep the following information about each cub in their group:



Personal Details Form

To ensure our records are up to date, please fill out all of the information below. Without a completed form, your child will not be able to participate in meetings/activities.

Personal Details	
Name:	
Date of Birth:	
Address:	
Gender:	
School:	
Telephone Number:	
Date Joined:	

▲ **Figure 9.7** Enrolment form

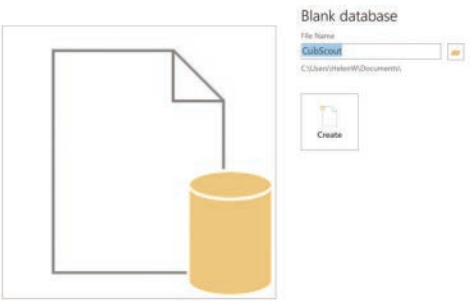
Define a single-table database from given data storage requirements and choose a suitable primary key

To create the cub scout database, open *Access* and select the `Blank database` template.



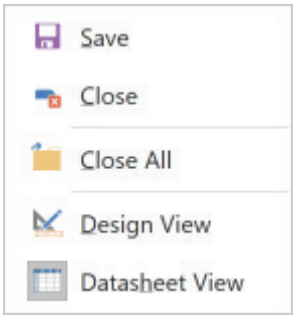
▲ **Figure 9.8** Blank database template

Then type the Filename `CubScout` and click the create button.



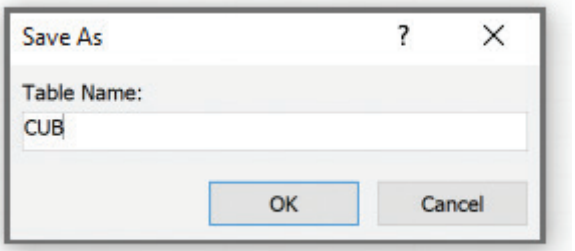
▲ **Figure 9.9** Creating the CubScout database

Select the table design view...



▲ **Figure 9.10** Design view

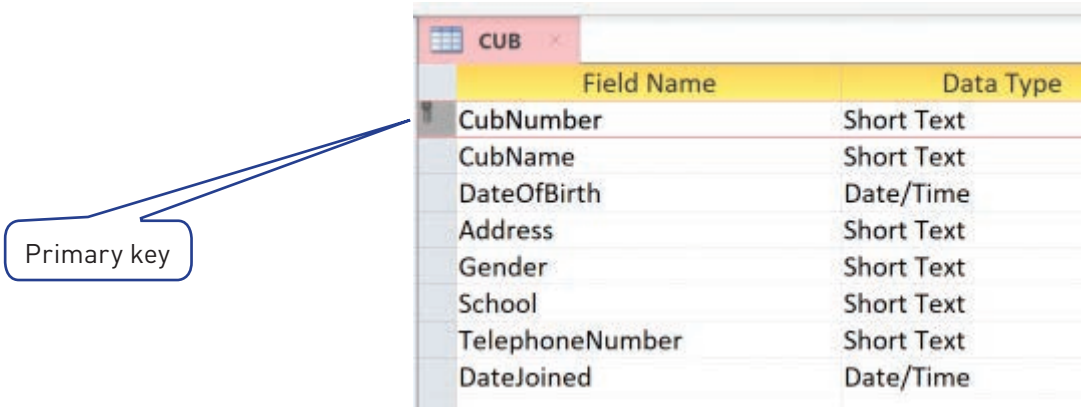
....and name the table **CUB**.



▲ **Figure 9.11** Naming the table

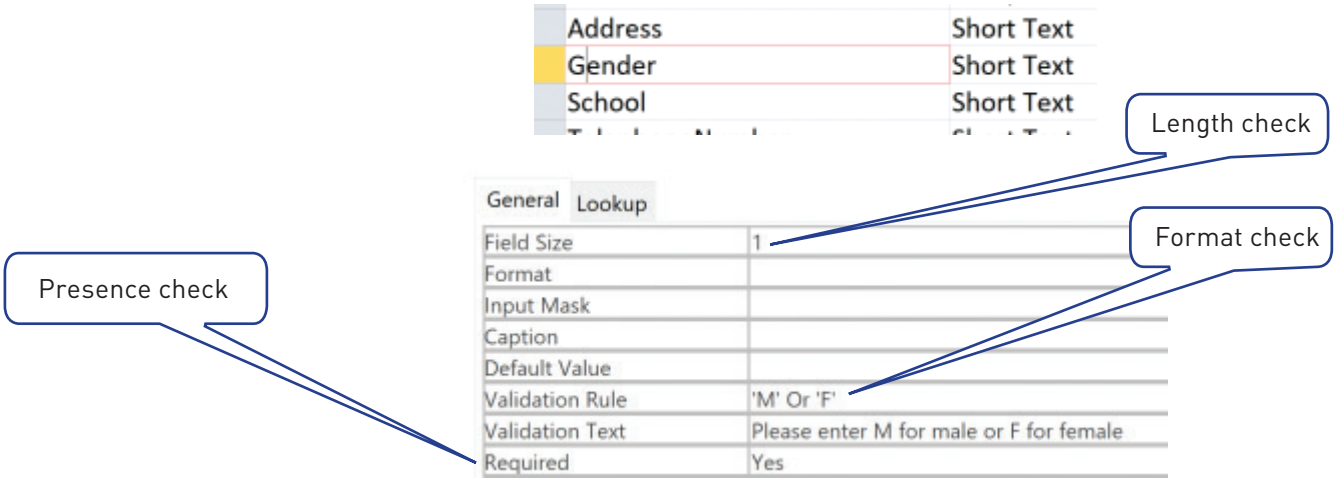
Set up the fields to match the data collection form in Figure 9.7 and include an extra field for a primary key.

Each field will require a meaningful name and a suitable data type must be selected.



▲ **Figure 9.12** Fields for the CUB table

Validation checks need to be built in for each field, for example the **Gender** field.



▲ **Figure 9.13** Validation rules for **Gender** field

Activity 9.7

1 **a** Set up a cub scout database including appropriate validation checks for each field.

Enter data for at least 10 records.

b The cub scout leader wants to put each cub into a group called a 'six'; each 'six' can have up to six cubs in it and is given a name for example red, yellow, blue and green.

Add a new text field called Six, and assign each cub to either a red, blue, yellow or green six.

i Write an SQL query to pick out any cubs in the red six.

ii Write an SQL query to pick out any cubs in the red six or the blue six.

iii Write an SQL query to count the number of cubs in the red six.

c The cub scout leader wants to calculate the number of badges that all the cubs have been awarded. Add a new integer field called *Badges*, enter the number of badges awarded to each cub.

Write an SQL query to count the number of badges awarded to the whole cub group.

Activity 9.8

1 A database of students is to be set up with the following fields:

- Family name
- Other names
- Student ID
- Date of Birth
- Date of Entry to School
- Current Class
- Current school year/grade
- Email address.

a Select a data type for each field.

b Which fields should be validated, and which fields should be verified?

c Decide the validation rules for those fields which should be validated.

d Which field would you choose for the primary key?

e Choose a suitable format for the student ID.

f Build a database with at least 10 records; include all your validation checks. Ensure there are at least 3 different classes and 2 different years/grades.

g Set up and test SQL scripts to:

i Display Other names, Family and Email address in alphabetical order of family name.

ii Select all the students from each class in alphabetical order.

iii Select all the students for each year/grade and print Other names, Family name and Date of Birth, grouping the students by class.

Link

For more on verification, see Chapter 7.

Extension

For those students interested in studying Computer Science at A Level, this is an extension of the use of SQL to build and modify a database as well as using SQL scripts to query an existing database.

AS and A Level covers relational databases that consist of more than one table.

Industry standard methods for building and modifying a database

Database Management Systems (DBMSs) use a **Data Definition Language (DDL)** to create, modify and remove the **data structures** that form a relational database. DDL statements are written as a script that uses syntax similar to a computer program.

DBMSs also use a **Data Manipulation Language (DML)** to add, modify, delete and retrieve the **data** stored in a relational database. DML statements are written in a script that is similar to a computer program.

These languages have different functions: DDL is used for working on the relational database structure, whereas DML is used to work with the data stored in the relational database.

Most DBMSs use **Structured Query Language (SQL)** for both data definition (DDL) and data manipulation (DML). SQL was developed in the 1970s and since then it has been adopted as an industry standard.

We have already covered some SQL commands used to manipulate data. These are the SQL (DDL) commands used to set up a database.

SQL (DDL) Command	Description
`CREATE DATABASE`	Creates a database
`CREATE TABLE`	Creates a table definition
`ALTER Table`	Alters a table definition
`PRIMARY KEY`	Adds a primary key to a table

These are the data types used for attributes (fields) in SQL.

Data Types for Attributes	Description
`CHAR(n)`	Fixed length text `n` characters
`VARCHAR(n)`	Variable length text maximum `n` characters
`BOOLEAN`	True or False SQL uses the integers 1 and 0
`INTEGER`	Whole number
`REAL`	Number with decimal places
`DATE`	A date usually formatted as YYYY-MM-DD
`TIME`	A time usually formatted as HH:MM:SS

Extension Activity

Use SQL (DDL) commands to set up the PATIENT database.

This SQL script could have been used to create the Cub Scout database.

```
CREATE DATABASE CubScout
CREATE TABLE CUB(
    CubName VARCHAR(30),
    DateOfBirth DATE,
    Address VARCHAR(40),
    Gender CHAR(1),
    School VARCHAR(30),
    TelephoneNumber, CHAR(14)
    DateJoined DATE);
ALTER TABLE CUB ADD PRIMARY KEY
(CubNumber);
```

In this chapter, you have learnt about:

- ✔ define a single-table database from a given set of requirements
- ✔ suggest and use suitable data types for fields
- ✔ choose an appropriate primary key for a record
- ✔ understand SQL scripts used to query a single-table database.

Key terms used throughout this chapter

database – a persistent structured collection of data that allows people to extract information in a way that meets their needs

single-table database – a database contains only one table

table – a collection of related records in a database

record – a collection of fields that describe one item

field – a database table

data type – a classification of how data is stored and displayed, and of which operations that can be performed on the stored value

primary key – a field in a database that uniquely identifies a record

Structured Query Language (SQL) – the standard query language for writing scripts to obtain useful information from a relational database.

SQL scripts – a list of SQL commands that perform a given task, often stored in a file so the script can be reused

SELECT – an SQL command that fetches specified fields (columns) from a table

FROM – an SQL command that identifies the table to use

WHERE – an SQL command to include only those records (rows) in a query that match a given condition

ORDER BY – an SQL command that sorts the results from a query by a given column either alphabetically or numerically

SUM – an SQL command that returns the sum of all the values in a field (column); used with SELECT

COUNT – an SQL command that counts the number of records (rows) in which the field (column) matches a specified condition; used with SELECT

Exam-style questions

1 A motor car manufacturer offers various combinations of:

- seat colours
- seat materials
- car paint colours.

A table, CAR, was set up as single-table database to help customers choose which seat and paint combinations were possible. (NOTE: N = no, not a possible combination, Y = yes, combination is possible)

Seat materials and colours

Paint colours

Code	Cloth	Leather	Seat colour	*White*	*Red*	*Black*	*Blue*	*Green*	*Silver*	*Grey*
CB	Y	N	black	Y	Y	Y	Y	Y	Y	Y
LB	N	Y	black	N	Y	N	N	N	Y	Y
CC	Y	N	cream	N	Y	Y	Y	N	N	N
LC	N	Y	cream	N	Y	Y	Y	N	N	Y
CG	Y	N	grey	N	Y	Y	Y	Y	Y	N
LG	N	Y	grey	N	Y	N	Y	N	Y	Y
CR	Y	N	red	Y	N	Y	N	N	Y	Y
LR	N	Y	red	Y	N	Y	N	N	Y	Y
CB	Y	N	blue	N	N	N	Y	N	N	N
LB	N	Y	blue	N	N	Y	Y	Y	N	N

a **i** State the number of records shown in the table. [1]

ii State the number of fields shown in the table. [1]

iii State, with a reason, if any of the fields are suitable to use as primary key. [2]

b The following SQL command was used:

```
SELECT Code
FROM CAR
WHERE SeatColour = 'Red' AND ('White' OR
'silver');
```

Show what will be displayed. [2]

c A customer wanted to know all the possible combinations for a car with leather seats and either silver or grey paint colour. Complete the SQL command required.

```
SELECT *
FROM ……
WHERE …… leather AND (silver …… grey);
```

[2]

2 A company that sells bicycles keeps records of the items in stock in a table, CYCLE, using a single-table database. For each model of bicycle, the following data is kept:

- model number, for example: BY00007
- description of model, for example: lady's shopper
- colour of frame, for example: gold and black

- size of wheels, for example: 700 mm
- price of model in $, for example: $309.50
- still being manufactured: Yes or No
- number in stock.

a **i** State, with a reason for your choice, the data type you would choose for each field in the table. [7]

ii Identify, with a reason for each, the validation required for each of these fields:
- model number
- price
- number in stock. [3]

iii State, with a reason, the field that would be suitable to use as primary key. [2]

b SQL commands are used to extract data from the table. Explain, using examples from your table, how each of these commands could be used:
- **`SELECT`**
- **`FROM`**
- **`WHERE`**
- **`SUM`**
- **`ORDER BY`** [10]

3 A database table, PERFORMANCE, is used to keep a record of the performances at a local theatre.

Show Number	Type	Title	Date	Sold Out
SN091	Comedy	An Evening at Home	01-Sep	Yes
SN102	Drama	Old Places	02-Oct	No
SN113	Jazz	Acoustic Evening	03-Nov	No
SN124	Classical	Mozart Evening	04-Dec	Yes
SN021	Classical	Bach Favourites	01-Feb	Yes
SN032	Jazz	30 Years of Jazz	02-Mar	Yes
SN043	Comedy	Street Night	03-Apr	No
SN054	Comedy	Hoot	04-May	No

a State the number of fields and records in the table.

Fields .....

Records ..... [2]

b Give **two** validation checks that could be performed on the Show Number field.

Validation check 1 .....

.....

Validation check 2 .....

..... [2]

Cambridge IGCSE Computer Science (0478) Paper 21 Q6 a), b), May/June 2018